

ELLEN ZHENG

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EDUCATION

Boston University, College of Engineering | Boston, MA

Expected May 2027

B.S. in Computer Engineering | Concentration: Machine Learning | Minor: Innovation & Entrepreneurship

Dean's List, **GPA: 3.95**

- **Coursework:** Machine Learning, Deep Learning, Robot Learning, Robotics & Autonomous Systems, Cyber-Physical Systems, Smart & Connected Systems, Algorithms & Data Structures, Software Engineering, Computer Architecture, Probability & Statistics

RESEARCH

Kulis Lab (Prof. Brian Kulis) | *ML and Robotics Researcher*

Sep 2026 – Present

- Studying how learned **multi-robot coordination** degrades under mid-episode agent failure, and whether failure-aware training yields graceful degradation without simulating every failure
- Establishing **MAPPO** baselines in VMAS, with planned hardware validation on LIMO robots

BU Robotics Lab (Prof. Roberto Tron) | *Computer Vision Researcher*

Sep 2025 – Present

- Built **ROS 2** depth pipeline (**ZoeDepth**, **PyTorch**) for collision avoidance on a LIMO robot; validated against tape-measured ground truth (0.3-1.5 m), where a quadratic refit cut depth error from **13.5 cm to 1 cm**
- Implemented a **Control Barrier Function** safety filter (closed-form QP, 20 Hz) with a bearing-weighted constraint limiting velocity toward obstacles, generating haptic force feedback for **shared autonomy**
- Integrated Geomagic Touch teleoperation by patching its **C++** driver to apply CBF forces each cycle; built **RViz** and **OpenCV** nodes for live debugging; presenting at UROP Symposium, publication planned

Multimedia Communications Lab | *Systems Researcher*

Sep 2024 – May 2025

- Won **Climate Action Prize** for an IoT sensing system deployed across 8 campus buildings; built analytics platform ingesting millions of time-series sensor readings (**Python/Pandas, MongoDB, Streamlit**)

PROJECTS

Neurobionics: EMG/EEG-Driven Rehabilitation Glove (ECE Senior Design, Project Owner)

Sep 2026 – Present

- Leading a **6-person** CompE/EE/BME team building an EMG-primary ML pipeline that decodes neuromuscular signals into motorized joint actuation; own system architecture, signal processing, and model selection

Imitation-Learning Driving Policy (EC518 Robot Learning)

Sep 2026 – Present

- Built a **behavior cloning** pipeline in **CARLA**: collected expert driving data, trained a **CNN** policy on camera input, and applied **DAGger** to correct compounding errors for a Town02 driving competition

Amyloid PET Centiloid Prediction (MedAI Hackathon)

Apr 2026

- Built a **3D ResNet-18** with FiLM tracer conditioning at every residual stage to predict amyloid burden from 3D brain PET scans across 4 radiotracers, trained on BU SCC GPUs
- Achieved the **best held-out test MAE of 12 teams**, cutting error 56% vs. baseline (12.5 vs. 28.1 Centiloid)

vLLM Inference Benchmarking (Red Hat vLLM/LLM-D Hackathon, Track 6)

Apr 2026

- Built a **vLLM** benchmarking and **regression detection** pipeline (GuideLLM, **Python**) that compares FP16 vs. FP8 runs and flags **TTFT** and throughput regressions against policy thresholds via CI exit codes
- Profiled FP8 Qwen3-8B at **~974 output tokens/s** and 7.5 req/s (concurrency 32) throughput vs. latency across chat and code

T-CRL: Temporal Causal Representation Learning (EC523 Deep Learning)

Spring 2026

- Built **PyTorch** pipeline predicting 5 mental health outcomes from smartphone sensor data (GLOBEM: 657 users, 3,390 features) with a temporal conv encoder, β -VAE, and a gate treating missingness as signal
- Reached **0.78-0.87 AUC-ROC** vs. 0.55-0.75 published benchmarks; gate beat no-gate **ablation** on 4/5 targets

EXPERIENCE

Capital One | *Business Analyst Intern* | McLean, VA

Jun – Aug 2026

- Scored data sources on coverage, freshness, and accuracy in **Snowflake SQL**, then formalized a scoring function and two-tier routing hierarchy for credit card application prefill, projected to lift completion **3-8%**

ReCo Digital | *AI Product Intern* | Sydney, Australia

Feb – Jun 2025

- Built a receipt analytics MVP in 4 weeks (**Python/Flask, Supabase/PostgreSQL**, OpenAI APIs, Gemini APIs) with **98%** extraction accuracy; cut page load times **40%** through query optimization

SKILLS

Robotics: ROS 2, URDF, RViz, OpenCV, CARLA, VMAS, PettingZoo, control barrier functions, camera calibration

ML: PyTorch, TorchRL, scikit-learn, NumPy, Pandas; DL, RL, generative models, imitation learning, computer vision, transformers

Languages & Tools: Python, C++, C, SQL, MATLAB, Verilog, React, Git, Linux, Docker, AWS, BU SCC (GPU cluster), vLLM

Methods: experiment design, ablations, evaluation metrics, statistical testing, regression